

CIVL6415, Semester 2, 2024

TRAFFIC ANALYSIS AND SIMULATION

MODULE 4

Macroscopic traffic models

Lecture Week 9

Dr. Mehmet Yildirimoglu
School of Civil Engineering
The University of Queensland
m.yildirimoglu@uq.edu.au

Flow conservation law

First-order models - Cell Transmission Model (CTM)

Second-order macroscopic models

Ramp Metering

The conservation law

- **An equation** relating point flows, i.e. $q(t, x)$, and densities, i.e. $k(t, x)$, in a road section:
 - **Theory of traffic flow** that can be used to predict traffic conditions in the network
- **No entering or exiting flow to the physical road section**

$$\frac{\partial q}{\partial x} + \frac{\partial k}{\partial t} = 0$$

“it ensures that flow and density variation in space and time is consistent with no entering / leaving traffic hypothesis.”

“entering and leaving traffic is equal in the space-time context”

Conservation law (visual)

entering and leaving traffic is equal

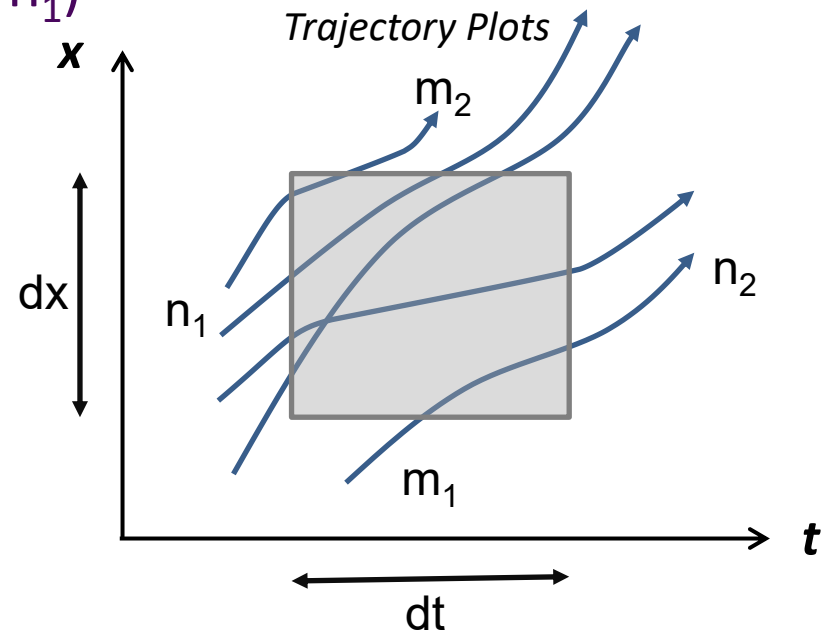
number of trajectories leaving: $(m_2 + n_2)$

number of trajectories entering: $(m_1 + n_1)$

$$(m_2 + n_2) = (m_1 + n_1)$$

$$(m_2 - m_1) + (n_2 - n_1) = 0$$

$$(3-1) + (2-4) = 0$$



Conservation law (visual)

entering and leaving traffic is equal

number of trajectories leaving: $(m_2 + n_2)$

number of trajectories entering: $(m_1 + n_1)$

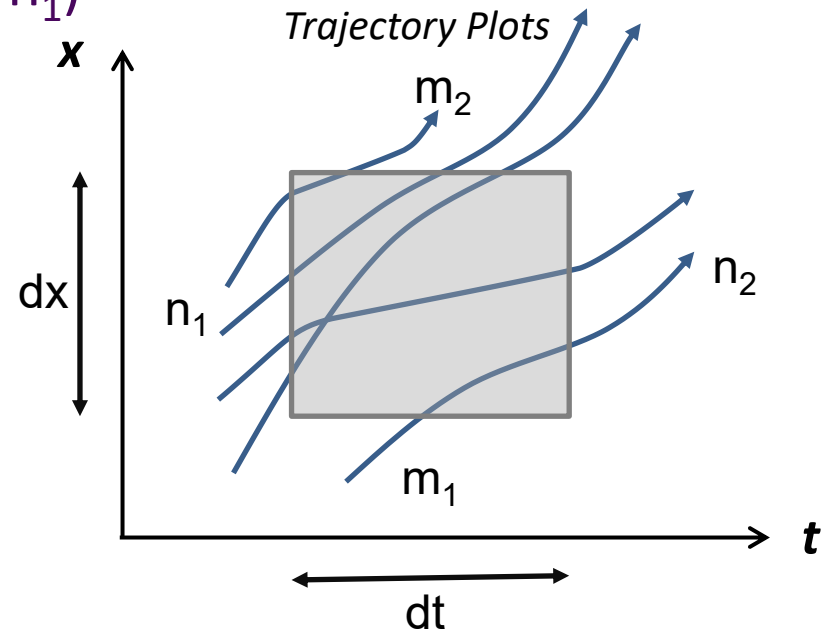
$$(m_2 - m_1) + (n_2 - n_1) = 0$$

$$(3-1) + (2-4) = 0$$

$$\frac{(m_2 - m_1)}{(dxdt)} + \frac{(n_2 - n_1)}{(dxdt)} = 0$$

$$\frac{q_2 - q_1}{dx} + \frac{k_2 - k_1}{dt} = 0$$

$$\frac{\partial q}{\partial x} + \frac{\partial k}{\partial t} = 0$$



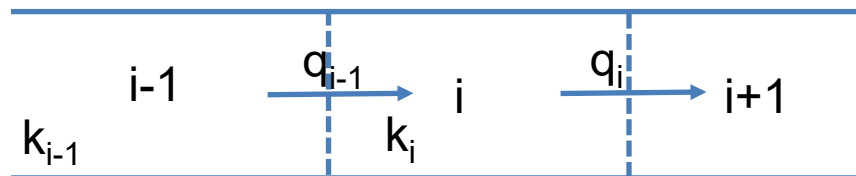
The conservation law

- Discretised formula
- T : length of time step, Δ_i : length of section i

$$\frac{\partial q}{\partial x} + \frac{\partial k}{\partial t} = 0$$

$$\frac{q_i(t) - q_{i-1}(t)}{\Delta_i} + \frac{k_i(t+1) - k_i(t)}{T} = 0$$

$$k_i(t+1) = k_i(t) + \frac{T}{\Delta_i} [q_{i-1}(t) - q_i(t)]$$



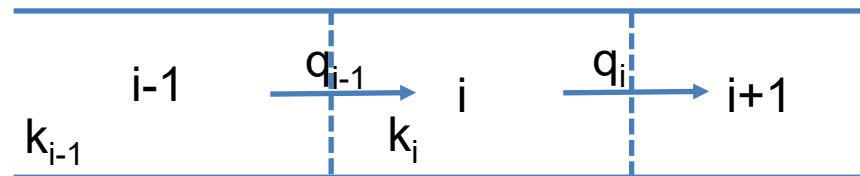
The conservation law

- Discretised formula
- T : length of time step, Δ_i : length of section i

$$k_i(t + 1) = k_i(t) + \frac{T}{\Delta_i} [q_{i-1}(t) - q_i(t)]$$

Diagram illustrating the discretised formula for the conservation law:

- $k_i(t + 1)$: Density at next time step
- $k_i(t)$: Density at current time step
- $q_{i-1}(t) - q_i(t)$: Inflow-Outflow



Flow conservation

Which one of the statements below is TRUE?

- A. The number of vehicles entering and leaving a physical roadway section in a given amount of time is always the same.
- B. Flow conservation law holds only if vehicles do not overtake each other
- C. Entering traffic is equal to leaving traffic in space-time context
- D. Flow conservation law cannot be used to predict traffic conditions
- E. None



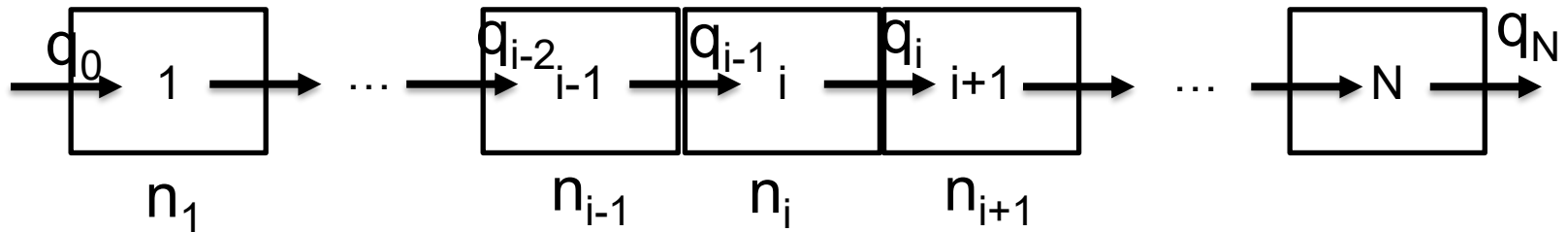
Cell Transmission Model (CTM)

- Analytical solution to kinematic wave model is exact
- Only applicable in relatively simple situations, e.g. with respect to upstream traffic demand, off-ramps and on-ramps, etc.
- What to do when demand on main-road and on-ramps is changing dynamically? Use numerical approximations!
- Discrete difference equations
- First order discrete Godunov approximation of LWR

Daganzo, Carlos F. "The cell transmission model: A dynamic representation of highway traffic consistent with the hydrodynamic theory." *Transportation Research Part B: Methodological* 28.4 (1994): 269-287.

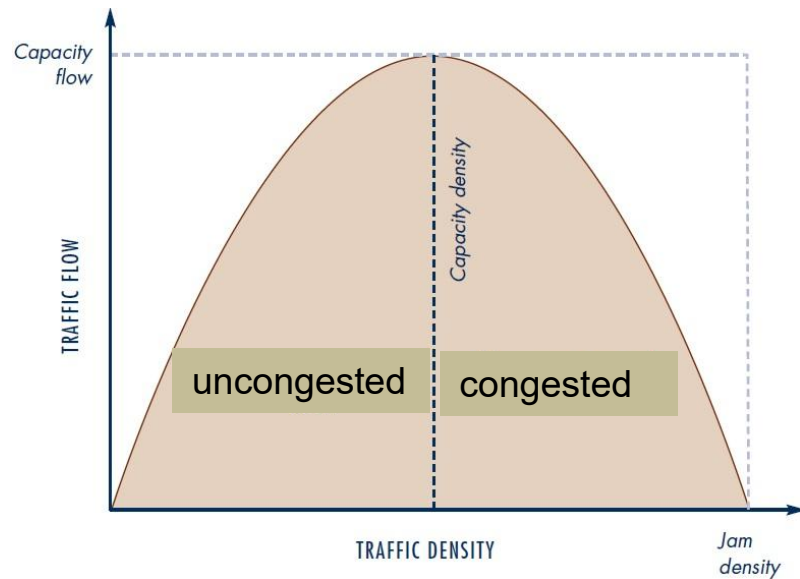
Daganzo, Carlos F. "The cell transmission model, part II: network traffic." *Transportation Research Part B: Methodological* 29.2 (1995): 79-93.

CTM assumptions



- Divide roadway into cells i , length dx
- Divide time into steps with length dt
- Cells are homogeneous
- Within the time step, traffic is stationary!

Fundamental diagram



The steady state relationship between flow and density

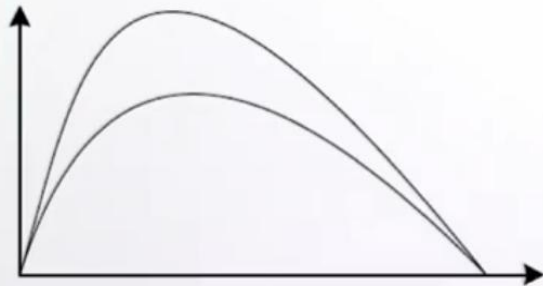
$$V(\rho) = v_f \cdot \exp\left(-\frac{1}{2}\left(\frac{\rho}{\rho_{cr}}\right)^2\right)$$

$$Q(\rho) = v_f \cdot \rho \cdot \exp\left(-\frac{1}{2}\left(\frac{\rho}{\rho_{cr}}\right)^2\right)$$

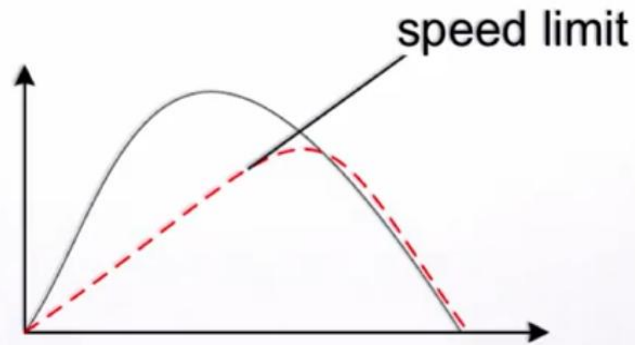
- **Macroscopic relationship:** $Q(\rho) = \rho \cdot V(\rho)$
- It does not violate the conservation equation!

FD - examples

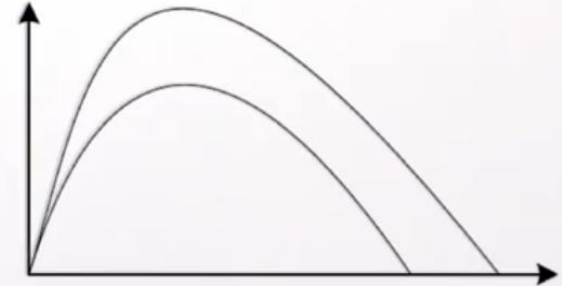
**Uphill,
curvature**



**Variable speed
limit (VSL)**

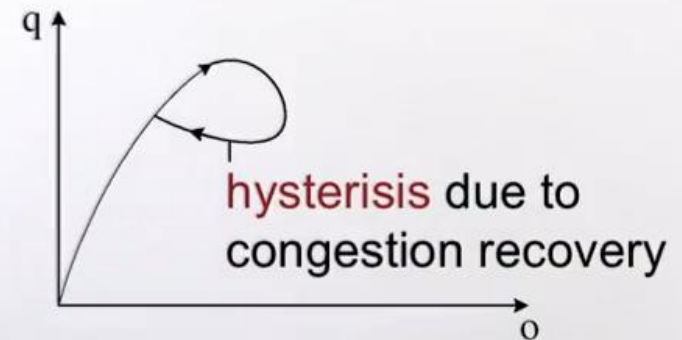
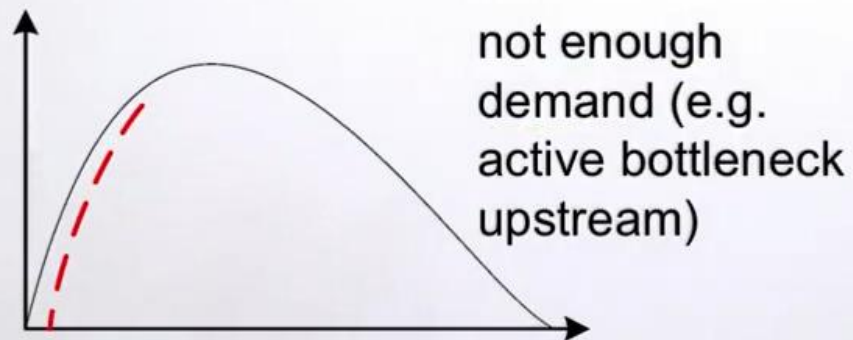
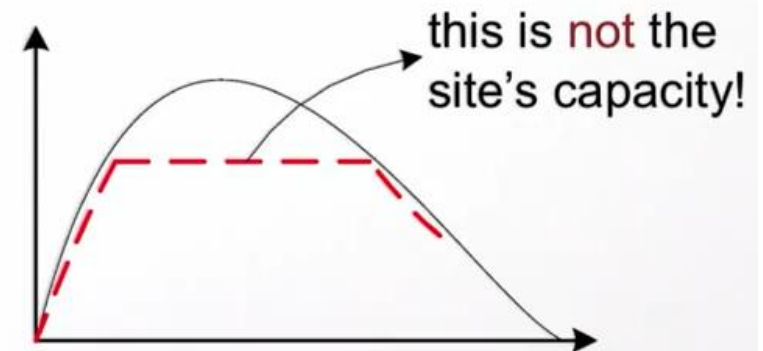
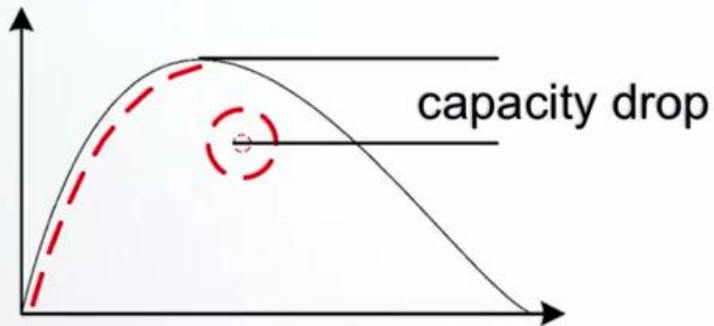


Lane drop

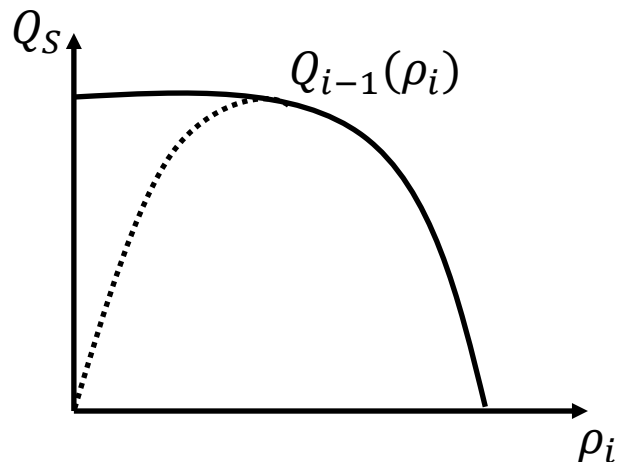
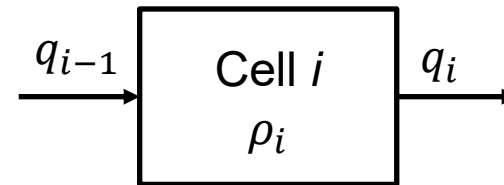
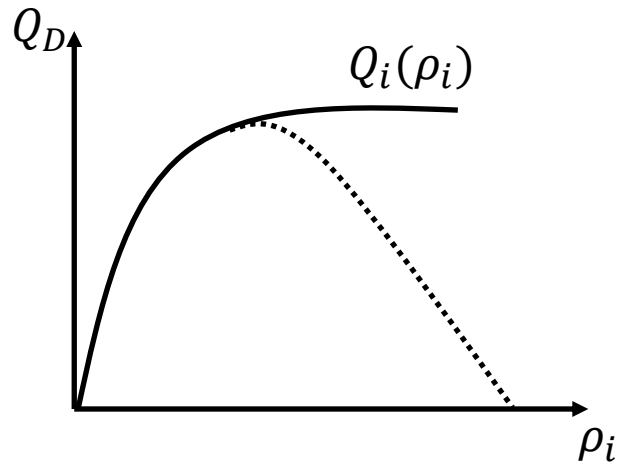


FD - limitations

Site measurements may deliver: - - - -

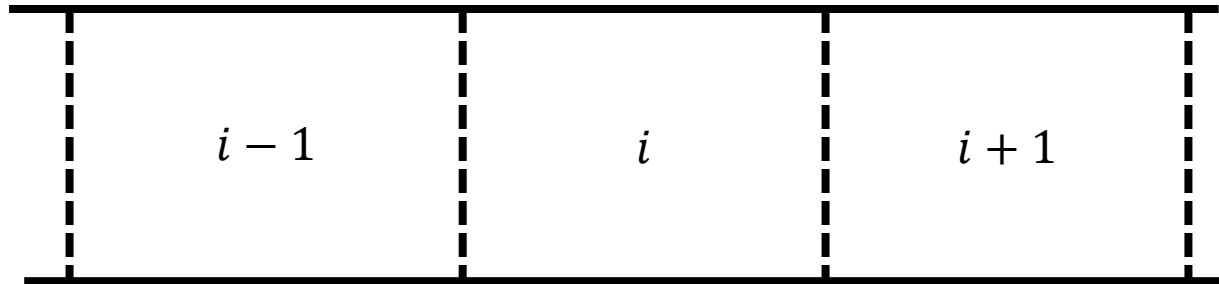


Demand / Supply FDs



The flows from one cell to the other restricted by

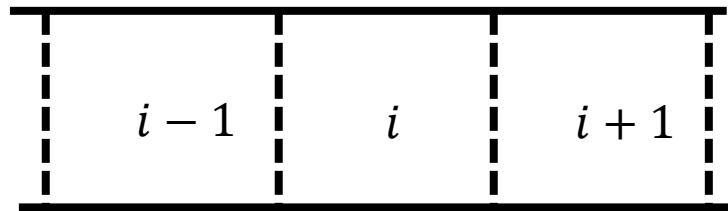
- Demand capacity
- Supply capacity



- **Density** $\rho_i(k)$: density in cell i at time step k or $t = kT$
- **Speed** $v_i(k)$: speed in cell i at time step k or $t = kT$
- **Flow** $q_i(k)$: number of vehicles leaving cell i between time step $k - 1$ and k or $[(k - 1)T, kT]$

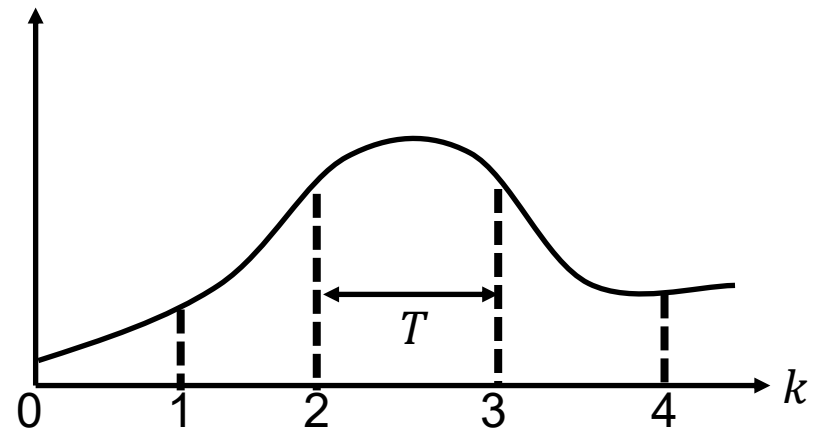
$$q_i(k) = \rho_i(k) \cdot v_i(k)$$

Spatial discretisation



$$\Delta_i \approx 10\text{-}500\text{m}$$

Temporal discretisation



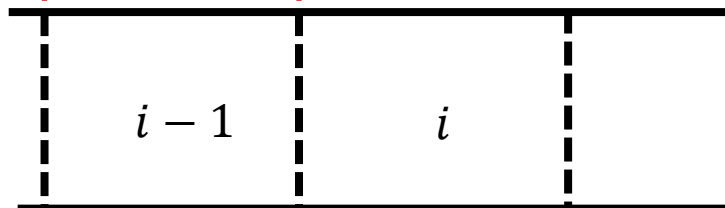
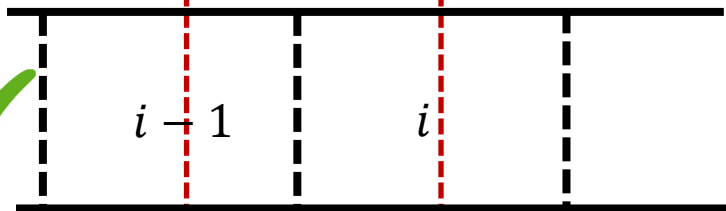
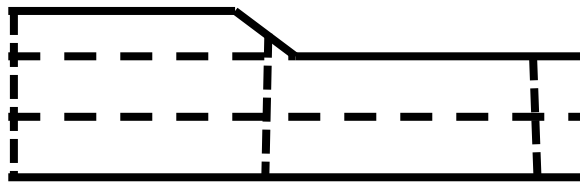
$$T \approx 0.2\text{-}15\text{s}$$

CFL conditions! $T < \frac{\Delta_i}{v_f} \quad \forall i$

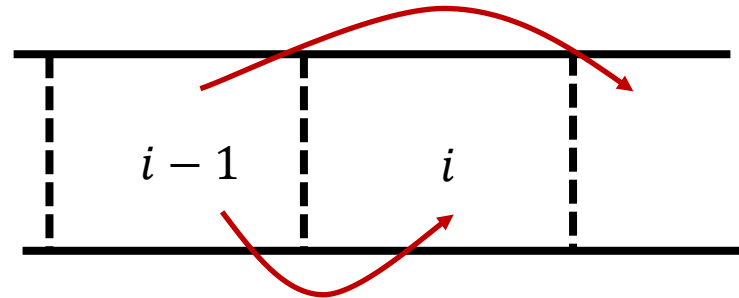
- In mathematics, the Courant–Friedrichs–Lewy (CFL) condition is a necessary condition for convergence while solving certain partial differential equations (usually hyperbolic PDEs) numerically.

Courant, R.; Friedrichs, K.; Lewy, H. (1928), "Über die partiellen Differenzengleichungen der mathematischen Physik", *Mathematische Annalen* (in German), 100 (1): 32–74.

Spatial discretisation



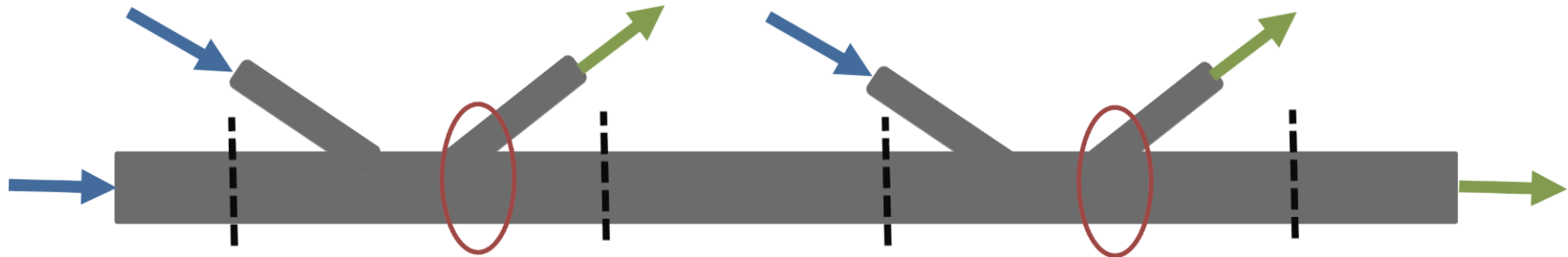
Temporal discretisation



CFL conditions! $T < \frac{\Delta i}{v_f} \quad \forall i$

Vehicles should not be able to traverse more than one cell in one time step

Freeway structure



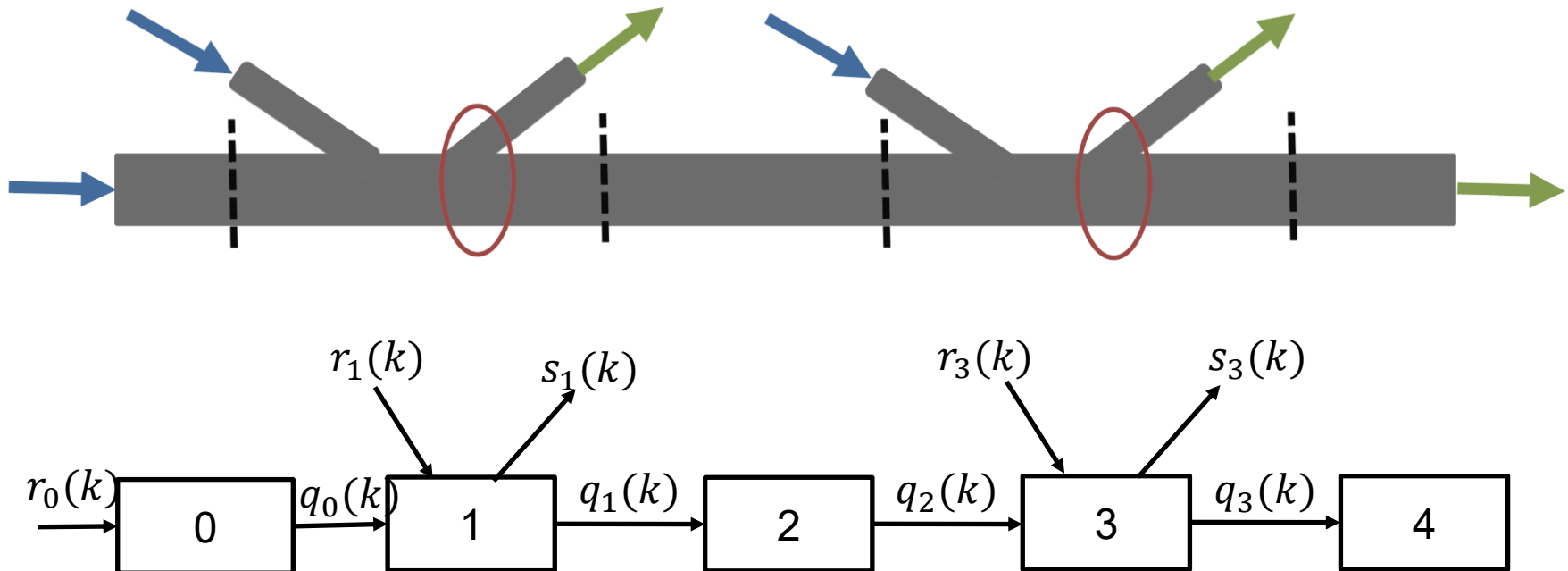
- Three requirements

Known demands

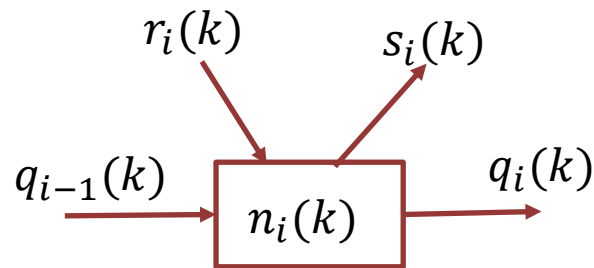
Known split ratios

Unobstructed outflows

Freeway structure



- A cell has at most one on-ramp and one off-ramp
- On-ramps have known inflows
- Off-ramps have known split-ratios



Split ratios

$$\beta_i(k) \in [0,1]$$
$$\bar{\beta}_i(k) = 1 - \beta_i(k)$$

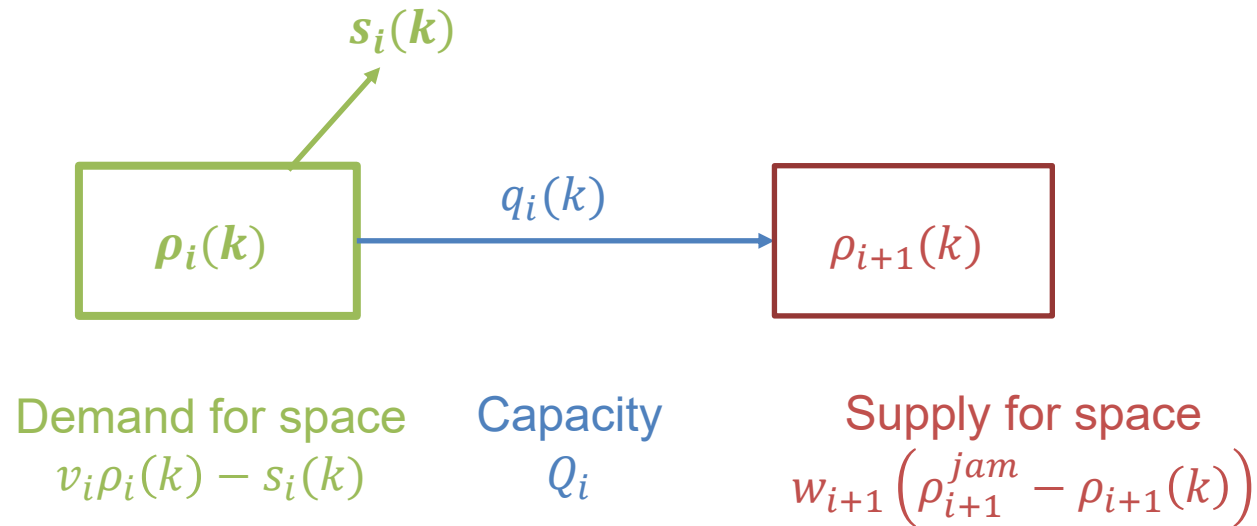
$$n_i(k+1) = n_i(k) + T [q_{i-1}(k) + r_i(k) - q_i(k) - s_i(k)]$$

$$\rho_i(k+1) = \rho_i(k) + \frac{T}{\Delta_i} [q_{i-1}(k) + r_i(k) - q_i(k) - s_i(k)]$$

Off-ramp flows

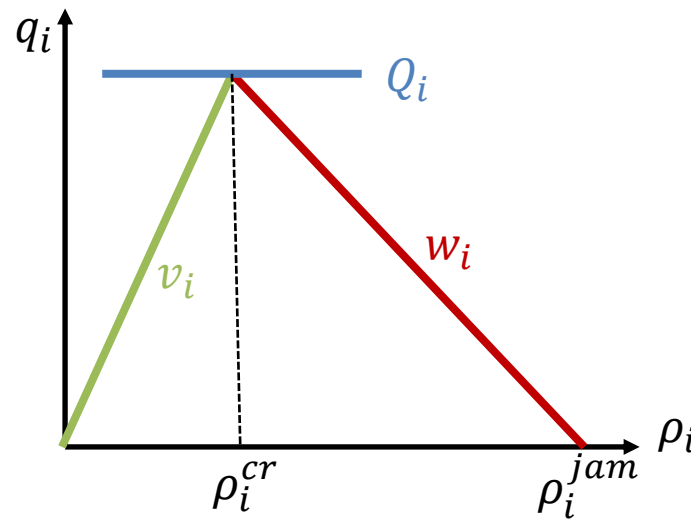
$$s_i(k) = \beta_i(k)[s_i(k) + q_i(k)]$$

$$s_i(k) = \frac{\beta_i(k)}{\bar{\beta}_i(k)} q_i(k)$$



$$q_i(k) = \min \left(\bar{\beta}_i(k) v_i \rho_i(k), w_{i+1} (\rho_{i+1}^{jam} - \rho_{i+1}(k)), Q_i \right)$$

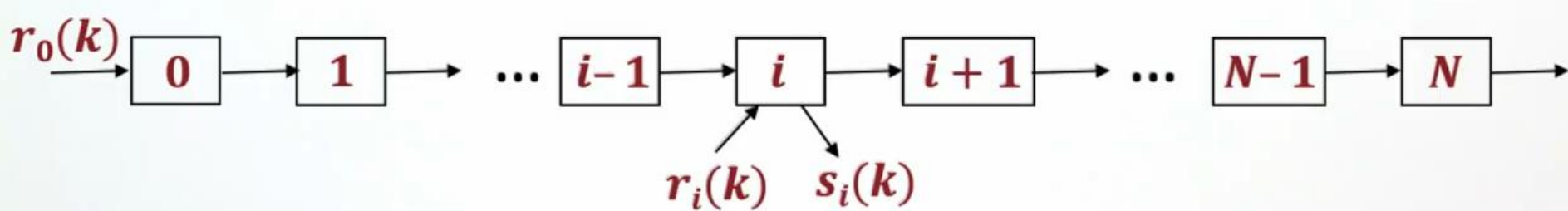
$$q_i(k) = \min \left(\bar{\beta}_i(k) v_i \rho_i(k), w_{i+1} \left(\rho_{i+1}^{jam} - \rho_{i+1}(k) \right), Q_i \right)$$



- Triangular FD implies, at $\rho_i = \rho_i^{cr}$

$$Q_i = \bar{\beta}_i(k) v_i \rho_i^{cr} = w_{i+1} (\rho_{i+1}^{jam} - \rho_i^{cr})$$

CTM – Putting it together



$$q_i(k) = \min \left(\bar{\beta}_i(k) v_i \rho_i(k), w_{i+1} \left(\rho_{i+1}^{jam} - \rho_{i+1}(k) \right), Q_i \right), \quad 0 \leq i \leq N-1$$

$$q_N(k) = \min(\bar{\beta}_N(k) v_N \rho_N(k), Q_N)$$

$$\rho_i(k+1) = \rho_i(k) + \frac{T}{\Delta_i} \left[q_{i-1}(k) + r_i(k) - \frac{q_i(k)}{\bar{\beta}_i(k)} \right], \quad 1 \leq i \leq N$$

$$\rho_0(k+1) = \rho_0(k) + \frac{T}{\Delta_0} \left[r_0(k) - \frac{q_0(k)}{\bar{\beta}_0(k)} \right]$$

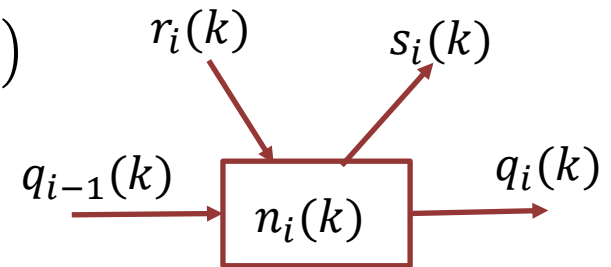
What if congested?

Density in cell i cannot exceed the jam density

1. Define a virtual queue for the on-ramp with $N_i(k)$ denoting the number of vehicles at time kT and $N_i(0)=0$.
2. At every time-step, calculate the two inflows that want to enter the cell as;

$$\overline{q_{i-1}}(k) = \min \left(\overline{\beta_{i-1}}(k) v_{i-1} \rho_{i-1}(k), w_i(k) \left(\rho_i^{jam} - \rho_i(k) \right), Q_{i-1} \right)$$
$$\bar{r}_i(k) = d_i(k) + \frac{N_i(k)}{T}$$

where $d_i(k)$ is the demand



3. If $\overline{q_{i-1}}(k) + \bar{r}_i(k) \leq w_i(k) \left(\rho_i^{jam} - \rho_i(k) \right)$, then you can apply these inflows, i.e. $q_{i-1}(k) = \overline{q_{i-1}}(k)$ and $r_i(k) = \bar{r}_i(k)$.

What if congested?

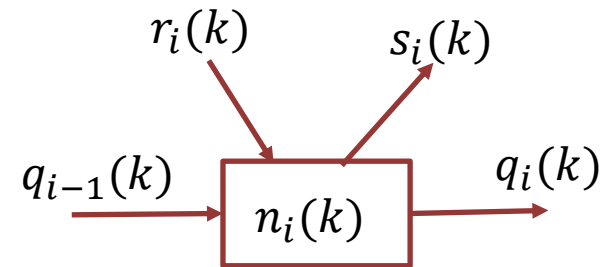
4. Else, the actual inflows are calculated

$$q_{i-1}(k) = \frac{\overline{q_{i-1}}(k)}{\overline{q_{i-1}}(k) + \overline{r_i}(k)} \cdot w_i(k) \left(\rho_i^{jam} - \rho_i(k) \right)$$

$$r_i(k) = \frac{\overline{r_i}(k)}{\overline{q_{i-1}}(k) + \overline{r_i}(k)} \cdot w_i(k) \left(\rho_i^{jam} - \rho_i(k) \right)$$

5. In both cases, the virtual queue is updated by

$$N_i(k+1) = N_i(k) + [d_i(k) - r_i(k)] \cdot T$$





Second-Order Models

CTM: a first-order traffic model

Issues with the first-order models

- Instantaneous change of speeds – Infinite acceleration
- Motion through shockwaves
- Driver anticipation not considered

Higher-order models

- Attempt to represent the acceleration of the traffic stream

Acceleration = $f(\text{traffic ahead, steady-state speed, reaction time, ...})$

Driver anticipation

Anticipation based on the density at $x + \Delta x$

$$v(x, t) = V(\rho(x + \Delta x, t))$$

$$\Delta x > 0$$

$$V(\rho) = v_f \cdot \exp\left(-\frac{1}{2} \left(\frac{\rho}{\rho_{cr}}\right)^2\right)$$

$v(x, t)$: Speed at location x and time t

Taylor series expansion;

$$v(x, t) = V(\rho(x, t)) + \left[\frac{\partial V}{\partial \rho} \frac{\partial \rho}{\partial x} \right]_{x,t} \cdot \Delta x$$

with $\Delta x \sim 1/\rho$ (1 vehicle ahead) – e.g., A/ρ (A vehicles ahead)

Assume an **anticipation constant**; $v = -A \partial V / \partial \rho > 0$: constant parameter

$$v(x, t) = V(\rho(x, t)) - \frac{v}{\rho(x, t)} \frac{\partial \rho}{\partial x}$$

Discretised version;

$$v_i(k) = V(\rho_i(k)) - \frac{v}{\Delta_i} \frac{\rho_{i+1}(k) - \rho_i(k)}{\rho_i(k) + \kappa}$$

A constant for the network
Gives the flexibility to calibrate

Reaction time

Reaction time, τ

$$v(x, t + \tau) = V(\rho(x + \Delta x, t))$$

$$v + \tau \cdot \frac{dv}{dt} = V(\rho) - \frac{v}{\rho} \frac{\partial \rho}{\partial x} \quad \leftarrow \text{Taylor series expansion}$$

with Chain rule

$$\frac{dv}{dt} = \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial t} = \frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x}$$

We obtain

$$\frac{\partial v}{\partial t} = \frac{dv}{dt} - v \frac{\partial v}{\partial x} = \left[V(\rho) - \frac{v}{\rho} \frac{\partial \rho}{\partial x} - v \right] \cdot \frac{1}{\tau} - v \frac{\partial v}{\partial x}$$

$$\text{Discretised version: } \frac{\partial v}{\partial t} = \frac{v_i(k+1) - v_i(k)}{T} \text{ and } \frac{\partial v}{\partial x} = \frac{v_i(k) - v_{i-1}(k)}{\Delta_i}$$

Dynamic equation

$$v_i(k + 1) = v_i(k)$$

$$+ \frac{T}{\tau} [V(\rho_i(k)) - v_i(k)]$$

Relaxation (to the steady-state)

$$+ \frac{T}{\Delta_i} v_i(k) [v_{i-1}(k) - v_i(k)]$$

Convection (tendency to travel with current speed)

$$- \frac{vT}{\tau \Delta_i} \frac{\rho_{i+1}(k) - \rho_i(k)}{\rho_i(k) + \kappa}$$

Anticipation (impact of density ahead)

Parameters to calibrate: τ , v , κ

Advantages:

- potential improvement of accuracy,
- fundamental diagram “noise” and capacity drop,
- traffic instability,
- congestion outflow

Limitations

- Inaccurate results with low traffic volumes
- lane-use restrictions, off-ramp spillback, dedicated lanes

No need to memorise the equations!!

Macroscopic models

Which one of the statements below is NOT TRUE?

- A. CTM is a numerical approximation of LWR theory
- B. CTM is a first-order model, as it does not account for the acceleration of the traffic stream
- C. Second-order models can produce capacity drop
- D. CTM accounts for driver anticipation
- E. None



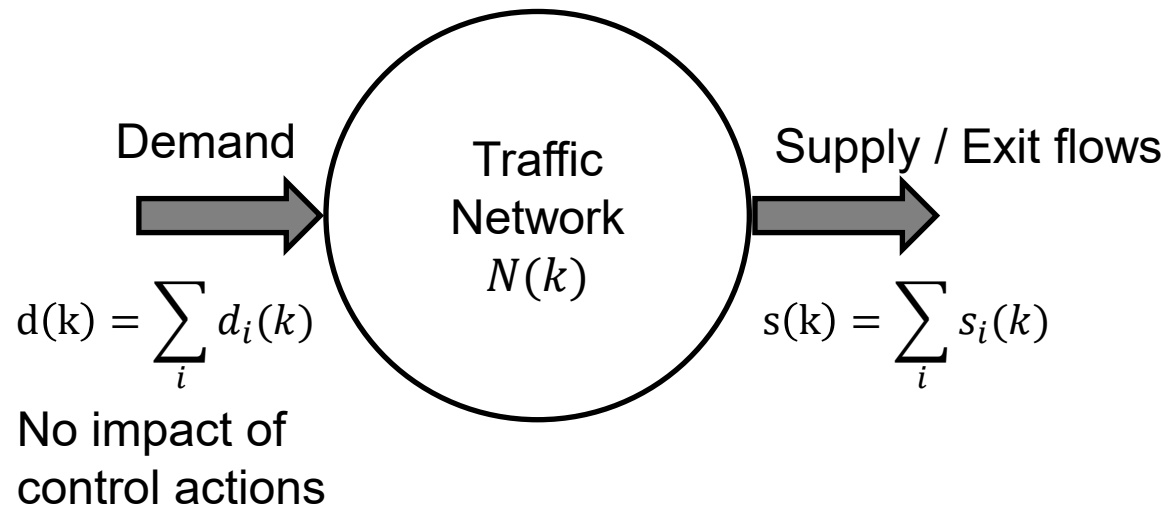
Freeway Control: Ramp Metering

- Originally...
 - sufficient capacity for virtually unlimited mobility (without the need for control)
- Ever increasing demand...
 - Congestions
 - Degraded infrastructure use
- Congestion: recurrent and non-recurrent
- Goal: Operate freeway networks optimally (as a **controllable** system)

- Ramp Metering
 - (direct impact on density $\Rightarrow$ congestion)
- Link Control
 - (variable speed limits, keep lane, lane control, congestion warning, tidal flow, environmental conditions: fog, ice, rain)
- Driver Information and Guidance
 - VMS Information
 - VMS Route Recommendation
 - Individual Route Guidance

<https://www.youtube.com/watch?v=8G7ViTTuwno>.

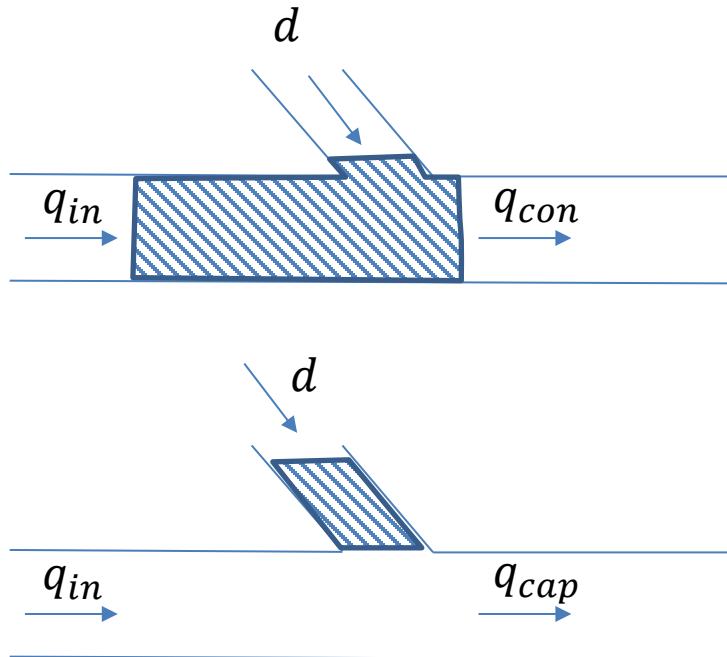
<https://youtu.be/QDMYODIgLcs?t=73>



- Total travel time or Total time spent $TTS = T \sum_{k=0}^K N(k)$
- How to minimise total time spent [veh.h]?
- $\min TTS, \max \sum_{k=0}^K (K - k) s(k)$

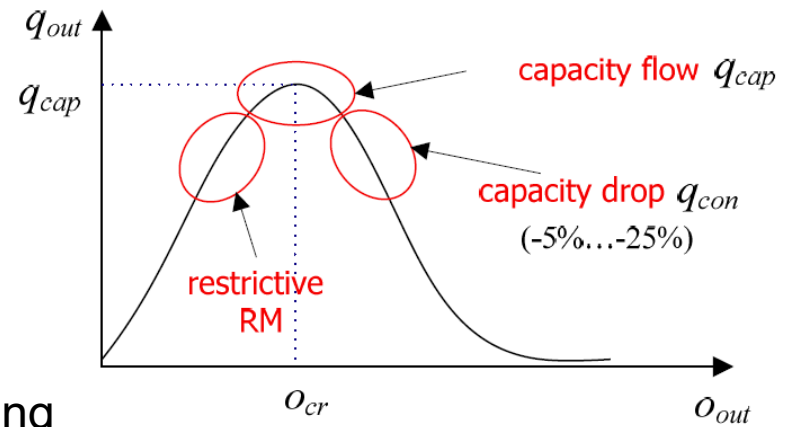
Ramp Metering

- Why useful? **1st answer**



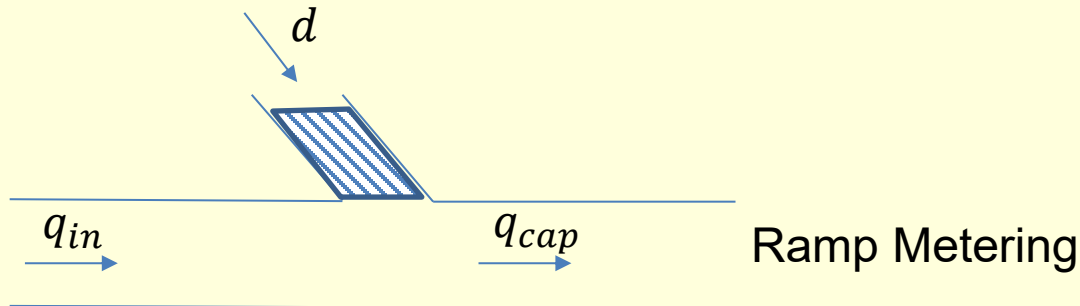
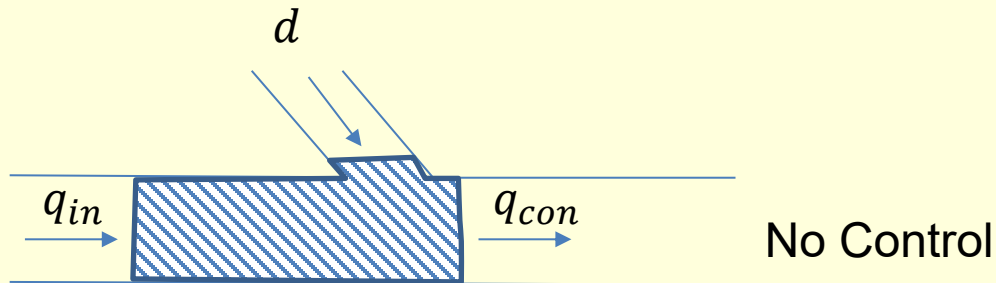
No Control

Ramp Metering



Ramp metering

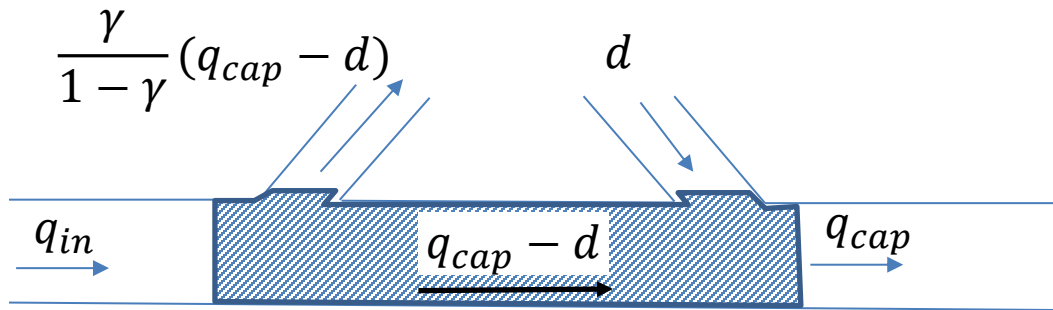
- Calculate travel time savings in terms of the variables given below.



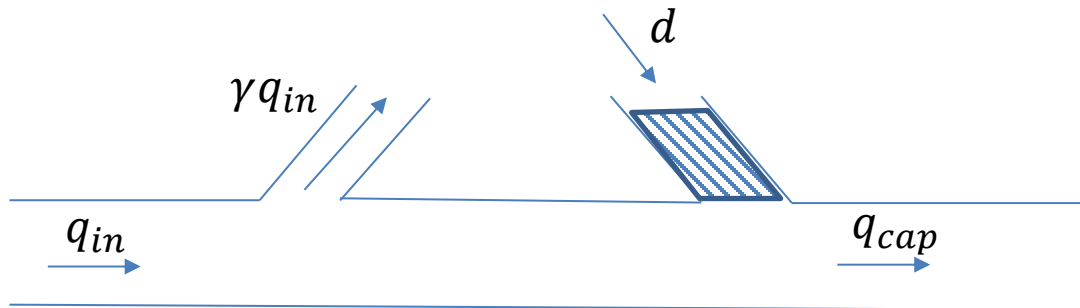
$$\Delta TTS = \frac{TTS^{nc} - TTS^{rm}}{TTS^{nc}} = ?$$

Ramp Metering

- Why useful? **2nd answer**



No Control



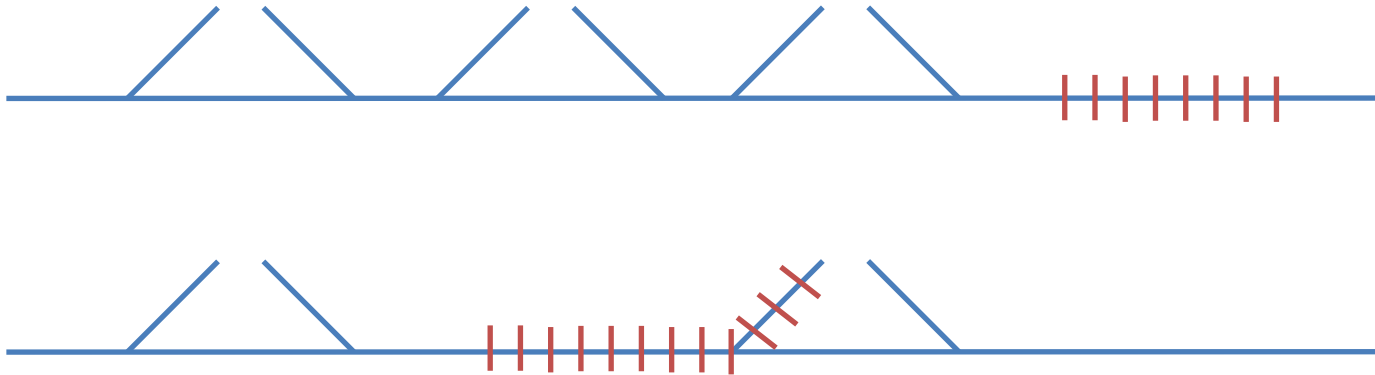
Ramp Metering

- Travel time savings

$$\Delta TTS = \frac{TTS^{nc} - TTS^{rm}}{TTS^{nc}} = \gamma$$

Ramp metering

- When not useful?

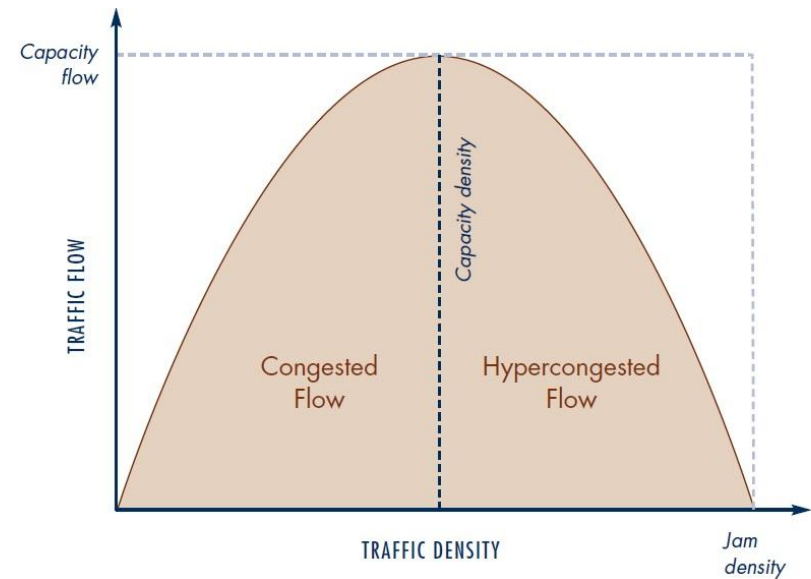
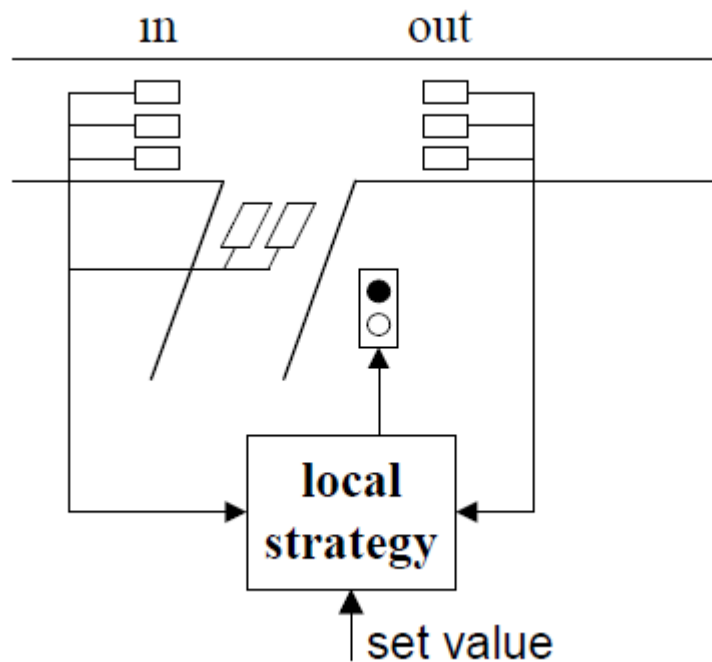


- Congestion further downstream
- Exit flow problems

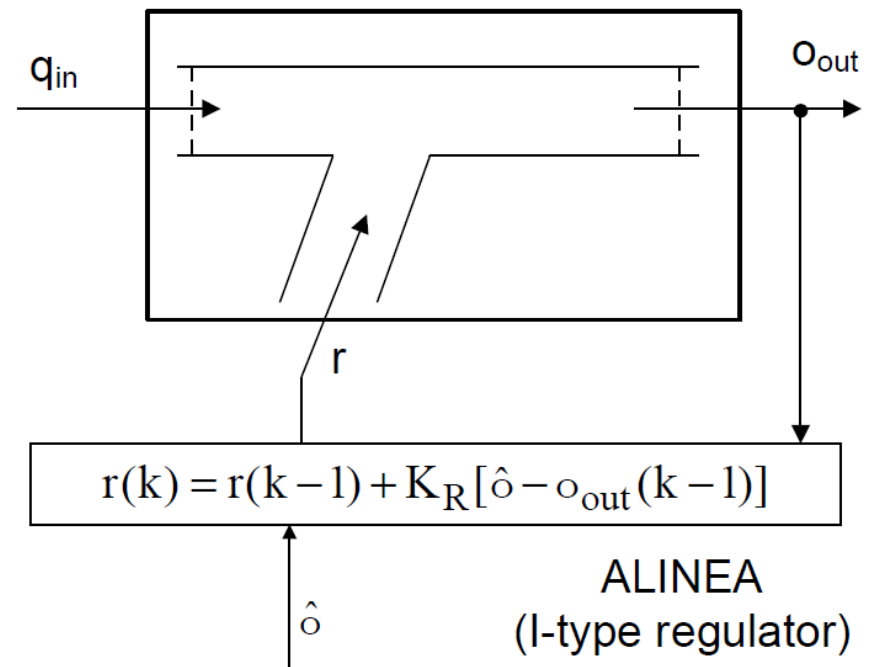
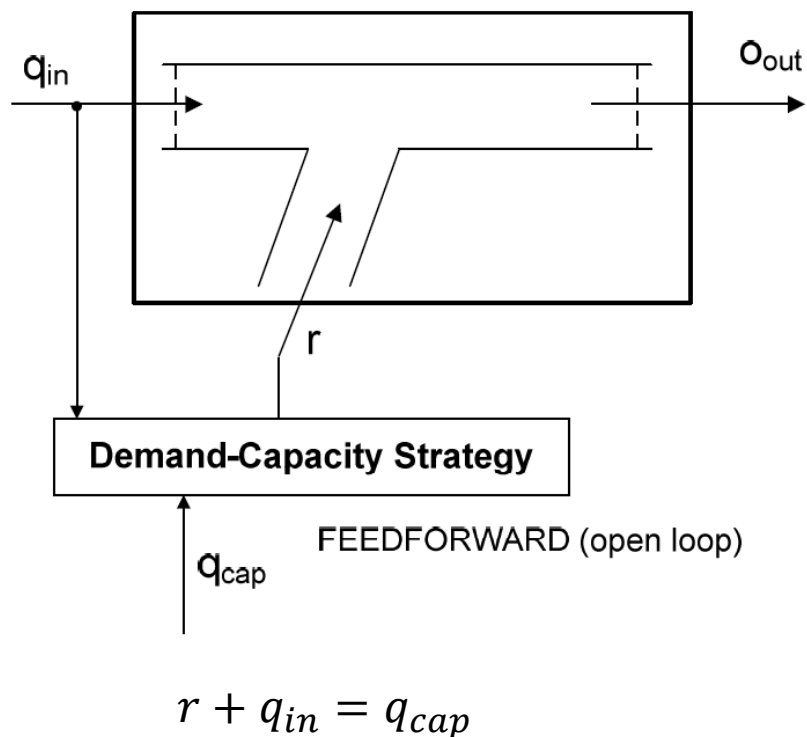
Note: On-ramp queue should not interfere with surface street traffic.

Regulator approaches

- Local ramp metering



Regulator approaches



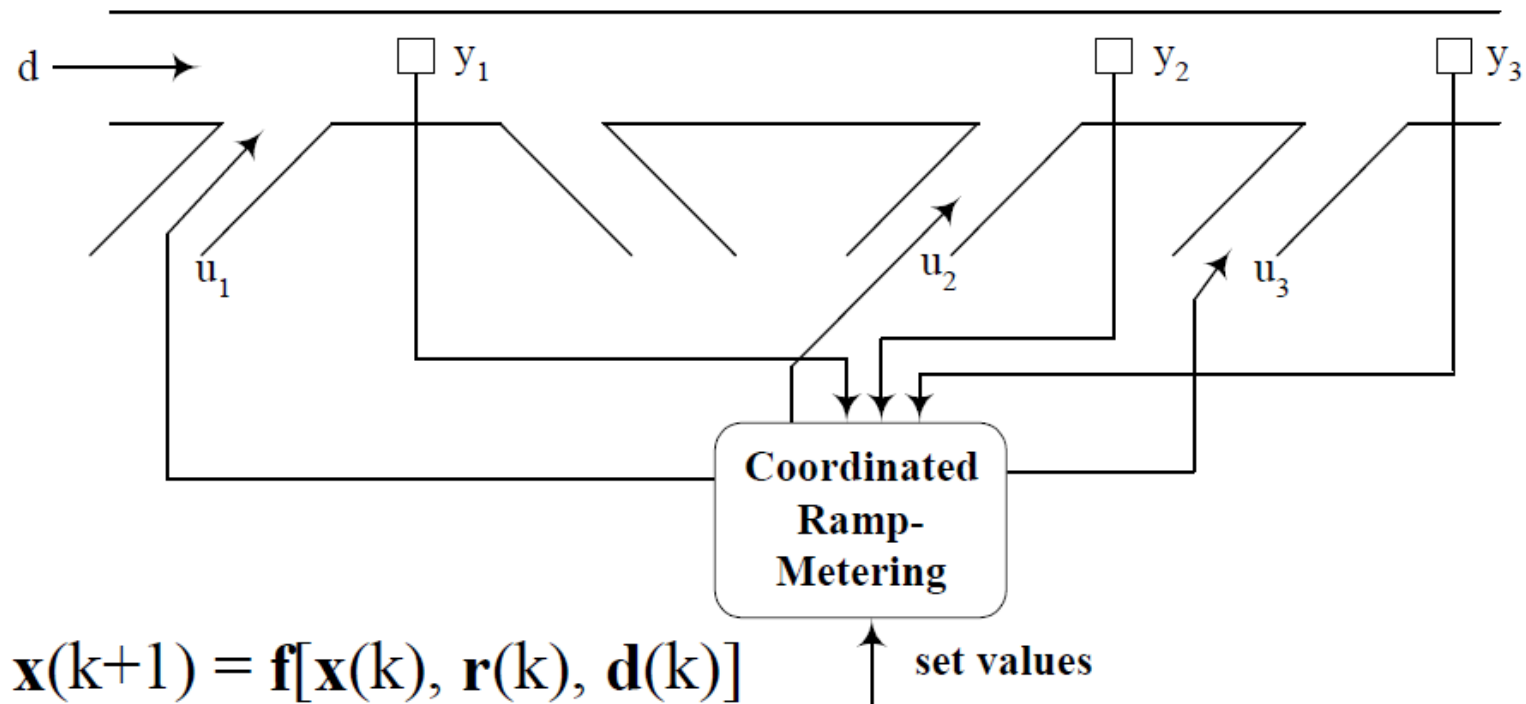
- **Traffic lights**
 - One car per green (adjust red phase wrt control flows)
 - Traffic cycles

- **Intermediate applications**
 - n cars at a time ($n=2,3,\dots$)
 - Short traffic cycles

- **Ramp capacity**
 - Usually exceeded
 - Important for overall efficiency

Coordinated ramp metering

- Improves efficiency and equity across the ramps
- Crucial when limited storage capacity



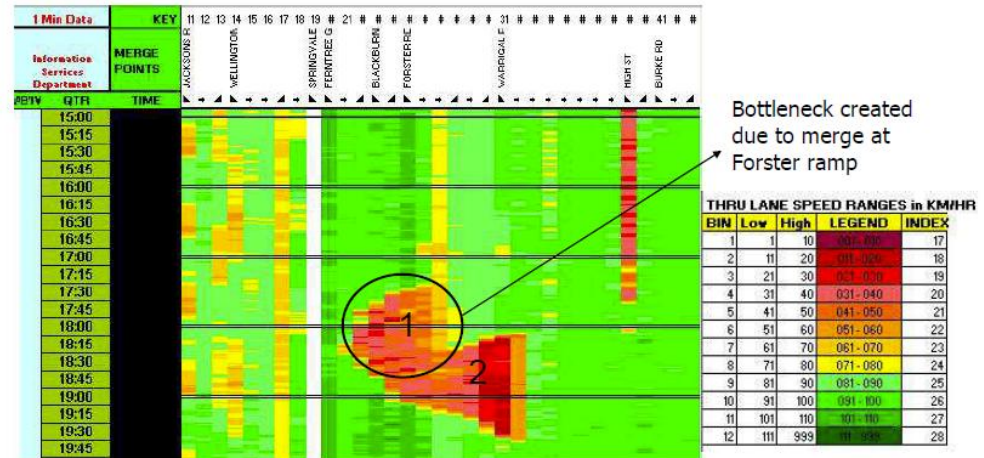
Coordinated ramp metering

- HERO (HEuristic Ramp metering cOordination)
- Monash freeway, Melbourne

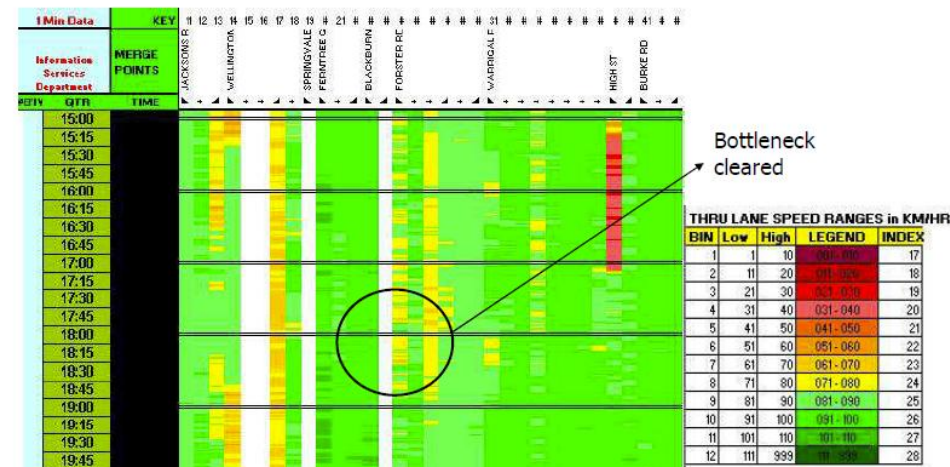


Papamichail, Ioannis, et al. "Heuristic ramp-metering coordination strategy implemented at monash freeway, australia." *Transportation Research Record* 2178.1 (2010): 10-20.

PM PEAK Typical day (No RM)



PM PEAK Typical day (ALINEA/HERO)



Next

This week

- Tutorial – Python

Computer exercise will build on this week's tutorial